

* Building blocks: Quantum bits

→ Find how these are realised in nature

Define:

(i) Quantum bits

(ii) Select a system in which the $|\psi\rangle$'s can evolve as desired (i.e. isolated)

(iii) Must be able to prepare $|\psi\rangle$'s in $|\psi_0\rangle = |\psi\rangle_{in}$

it must be able to measure $|\psi\rangle_{out}$

• Single nuclear spin can be used:

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$$

Can stay aligned with an external $\vec{B}$ for days.

→ Practically: Very hard to measure the orientation of a single nuclei.

* A Quantum computer has to be well isolated in order to retain its quantum properties

But:

its qubits have to be accessible so they can be manipulated to perform a computation and hence find a given result.

→ Idea: Not if we can build a Q. Comp but how well can we balance isolating a state → finding a way to measure / interact w/ this state.

* Study: Quantum noise/decoherence

T₁: time for which a system remains coherent.

coherent

T_{op} : time it takes to perform elementary unitary transformations (involves at least 2 qubits)

$$\lambda \equiv T_{op} / T_Q$$

System	T_Q	T_{op}	λ^{-1}
nuclear spin	$10^{-2} - 10^8$	$10^{-3} - 10^{-6}$	$10^5 - 10^{14}$
Elec. spin	10^{-3}	10^{-7}	10^4
Ion trap ($^{17}\text{F}^+$)	10^{-1}	10^{-14}	10^{13}
Electron-Au	10^{-8}	10^{-14}	10^6
Electron-GaAs	10^{-10}	10^{-13}	10^3
Quantum dot	10^{-6}	10^{-9}	10^3
Optical cavity	10^{-5}	10^{-14}	10^9
Microwave cavity	10^0	10^{-4}	10^4

$$(T_{op}) = T_{op},$$

• System is only together for T_Q , then hope $T_Q > T_{op}$

$$n_{op} \equiv \frac{T_Q}{T_{op}} = \lambda^{-1} \quad (\text{maximum \# of ops that can be made})$$

* Four - Basic Requirements for quant. Comp.

(i) Robustly represent quant. inf.

(ii) Perform a universal family of unitary trans.

(iii) Prepare a fiducial initial state

(iv) Measure the output result

(i) Robustly rep. quant. info:

(Quantum Computation) $\longleftrightarrow$ (Transformation of quant. states)

ex: $l = 3/2$, $|3/2, 3/2\rangle$, $|3/2, 1/2\rangle$, $|3/2, -1/2\rangle$, $|3/2, -3/2\rangle$

A spin $3/2$ particle ψ_{max} has 4 different states and can be used to represent two-qubits.

$$\bullet (|3/2, 3/2\rangle, |3/2, -3/2\rangle) \oplus (|3/2, 1/2\rangle, |3/2, -1/2\rangle)$$

* Set of accessible states should be finite.

• A state $|\psi\rangle = \sum_k c_k |k\rangle$

is not a good system as $\mathcal{H}$ is ∞ -dim

• Want symmetry to dictate the finiteness of the state space: minimizes decoherence

ex: Spin $1/2$ particle:

$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle$$

nearly ideal when there is no noise.

$\leadsto$ If the choice of rep. is poor, then decoherence will occur.

ex: (Square wells & qubits) $V = \begin{cases} 0, & 0 < x < L \\ \infty, & x \notin (0, L) \end{cases}$



$$i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V(x) \psi$$

$$\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V(x) \psi = E \psi$$

$$\partial_x^2 \psi = -\frac{2m}{\hbar^2} (E - V(x)) \psi$$

$$\omega^2 = \frac{2m}{\hbar^2}$$

$$\begin{cases} V_0: & \partial_x^2 \psi = -\frac{2m}{\hbar^2} E \psi \\ V: & \psi = 0 \end{cases}$$

$$\psi(x) = A \sin\left(\sqrt{\frac{2mE}{\hbar^2}} x\right) + B \cos\left(\sqrt{\frac{2mE}{\hbar^2}} x\right)$$

Match Boundary Conditions:

$$\psi(0) = 0 = B$$

$$\psi(L) = 0 = A \sin\left(\sqrt{\frac{2mE}{\hbar^2}} L\right)$$

$$\sqrt{\alpha} L = \pi n$$

$$\sqrt{\alpha} = \pi n / L$$

$$\psi(x) = \begin{cases} A \sin\left(\frac{\pi n}{L} x\right), & x \in (0, L) \\ 0 & x \notin (0, L) \end{cases}$$

$$\langle \psi_n, \psi_m \rangle = \int_0^L dx A^2 \sin\left(\frac{\pi n}{L} x\right) \sin\left(\frac{\pi m}{L} x\right)$$

$$= \frac{L}{2} A^2 = 1$$

$$A = \sqrt{2/L}$$

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{\pi n}{L} x\right) & x \in (0, L) \\ 0 & x \notin (0, L) \end{cases}$$

$$(2mE/\hbar^2) = \frac{\pi^2 n^2}{L^2} \Rightarrow E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

$$\text{Thus: } \langle x | \psi_n \rangle = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi n}{L} x\right)$$

$$|\psi_n(t)\rangle = e^{-iE_n t} |\psi_n\rangle$$

• Not sure how this is finite, Let $|\psi_1\rangle$ and $|\psi_2\rangle$ be any two states, then

$$|\psi(t)\rangle = a |\psi_1(t)\rangle + b |\psi_2(t)\rangle$$

$$|\psi(t)\rangle = a e^{-iE_1 t} |\psi_1\rangle + b e^{-iE_2 t} |\psi_2\rangle$$

$$= e^{-i(E_1/2)t} e^{-i(E_2/2)t} \left(a e^{iE_1/2t} e^{iE_2/2t} |\psi_1\rangle + b e^{iE_1/2t} e^{iE_2/2t} |\psi_2\rangle \right)$$

Define

$$\omega = (E_1 - E_2)/2$$

$$|\psi(t)\rangle = e^{-i(E_1+E_2)t/2} \left(a e^{-i\omega t} |\psi_1\rangle + b e^{i\omega t} |\psi_2\rangle \right)$$

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\downarrow$ $|\psi_1\rangle$ $\downarrow$ $|\psi_2\rangle$

Time evolution: $e^{-iHt} \rightarrow H = \hbar\omega Z$

• Perturb H :

$$V \rightarrow V + \delta V$$

$$\delta V(x) = -V_0(t) \frac{q\pi^2}{16L} \left(\frac{x}{L} - \frac{1}{2} \right)$$

$$V_{nm} \equiv \langle \psi_n | \delta V(x) | \psi_m \rangle \rightarrow H_1 = V_0(t) \hat{X}$$

→ This slight variation to the potential (i.e. interference) will have an overall effect on computational op. of qubits.

Performance of unitary transformations:

• Must be able to control the Hamiltonian to effect an arbitrary selection from a universal family of unitary transformations: (Quantum Gates)

ex: $H = P_x(t) \hat{X} + P_y(t) \hat{Y}$

• Manipulating P_x, P_y we can perform arbitrary single spin rotations

Thm: Any unitary ($UU^\dagger = I$) transform can be composed from single spin ops & controlled-NOT gates

$$U = V_1 \dots V_n$$

- Varying transformations can lead to decoherence

$$R(\epsilon) |n, l, m\rangle$$

leads to loss of quantum information, represented by the relative phases in a quantum state.

- It should include the back-action of the control system upon the quantum computer

• In practice: $P_x \rightarrow P_x(t) = \sum_k \omega_k(t) (a_k + a_k^\dagger)$

After interacting, the photon^k will carry information

↳ decoherence

* Completely unitary system will have no info loss.
